SuddWatch Phase 2: Pipeline Execution, Visual Diagnostics and Strategy-versus-Reality Audit

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***Abstract*—This report documents the execution of the SuddWatch Phase 1 preprocessing strategy on the same 23,305 × 26 county-window panel. A single scikit-learn pipeline (county-median imputation, a momentum feature, rainfall winsorisation and a column transformer) is fitted on training rows only and applied unchanged to the validation and test periods, producing 23 model-ready features. Eight diagnostic figures verify each step. A decision log compares all twenty Phase 1 commitments with what was implemented: sixteen ran as planned, three were adapted and one is deferred to the modelling stage. Reducing the feature set from 34 to 23 columns did not lose information: temporal cross-validated PR-AUC of a baseline classifier rose from 0.534 to 0.550. Two findings carry forward: the label sensitivity check kept the Phase 1 label, and exceedance rates rise steadily from 2022, a drift risk for modelling.**

***Index Terms*—data preprocessing, scikit-learn pipeline, data leakage, Sentinel-1, flood forecasting, South Sudan.**

I. Introduction

Phase 1 audited the SuddWatch panel and fixed a preprocessing strategy before any production code was fitted. Phase 2 executes that strategy in `phase2_implementation/notebooks/pipeline_execution.ipynb`, which imports the same modules (`src/transformers.py`, `src/pipeline_builder.py`) and starts by asserting that the raw panel still reproduces the Phase 1 audit exactly (23,305 rows, 26 columns, 1.46% missing cells, 19,708 complete rows). Section II describes the pipeline, Section III the eight diagnostic figures, Section IV the decision log, Section V the deviations and their rationale, and Section VI the output verification.

II. Pipeline Implementation

A. Architecture

Table I lists the steps in execution order. Every fitted statistic is learned from `X_train` only; validation and test are passed through `transform` and are never used to fit anything. The test set is not scored in this phase.

Table I
Pipeline Steps (Phase 1 Section in Brackets)

#	Step	Fitted on	Purpose
0	Temporal split + purge	—	Rows split by issue date before any fitting; 2 boundary windows purged (VI-A, VI-D).
1	`GroupMedianImputer`	`X_train`	County training medians + `_was_missing` flag for Sentinel-1 water features (IV-A/B).
2	`MomentumFeature`	stateless	`water_frac_delta` = `water_frac` − `water_frac_lag1` (Phase 1 proposal, IV-F).
3	`OutlierCapper(z)`, `OutlierCapper(iqr)`	`X_train`	Winsorise rainfall at training μ ± 3σ / 1.5×IQR fences (IV-C/D).
4	`ColumnTransformer`	`X_train`	log1p(100·x)+Std (water), log1p+Std (skewed), Std (symmetric), MinMax (bounded), one-hot state, sin/cos month; Phase 1 drop list applied (V-A/B, IV-F).
5	Estimator (later)	`X_train`	Logistic-regression baseline used here only to verify the preprocessing (Section VI).

B. Temporal Split

Rows without a label are excluded and the two issue windows before each split boundary are purged, because the target looks two windows ahead and would otherwise overlap the next split. Table II gives the resulting partitions.

Table II
Temporal Partitions After Purging

split	issue dates	rows	share	positives	pos. rate	Phase 1 share
train	2017-01-12 → 2023-12-07	15,424	73.9%	2,926	19.0%	73.5%
validation	2024-01-12 → 2025-06-11	2,792	13.4%	1,030	36.9%	13.9%
test	2025-07-17 → 2026-08-29	2,662	12.8%	980	36.8%	12.6%

C. Processed Output

The fitted pipeline maps the 25 raw input columns of each split to 23 numeric features: train 15,424 × 23, validation 2,792 × 23, test 2,662 × 23, with no missing values. The matrices, the feature names and the fitted pipeline (`preprocessor.joblib`) are saved in `data/processed/`.

III. Visual Diagnostics

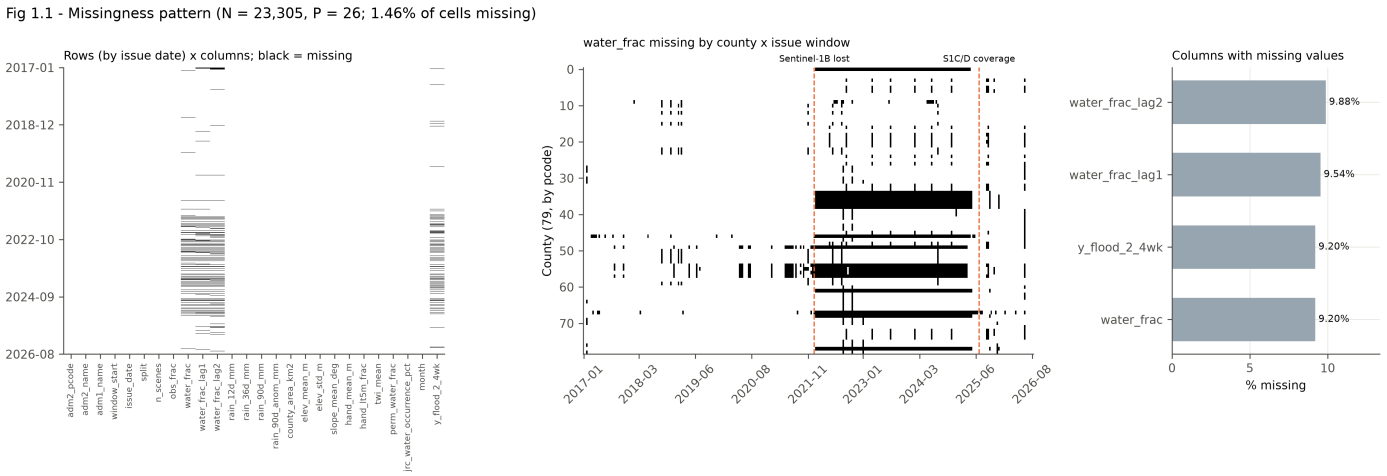


Fig. 1. Missingness pattern (assignment Fig. 1.1). Left: row × column matrix. Middle: `water_frac` gaps by county and issue window; the dense blocks between the Sentinel-1B loss and Sentinel-1C/D coverage are the MAR acquisition-plan gaps diagnosed in Phase 1. Right: missing percentage per column.

Fig. 1. Missingness is confined to the four Sentinel-1-derived columns and concentrated in a fixed set of counties between Dec 2021 and mid-2025, as diagnosed in Phase 1.

Fig. 2. The training set is imbalanced (19.0% positive), while validation and test are near 37% positive. Section V-C examines why.

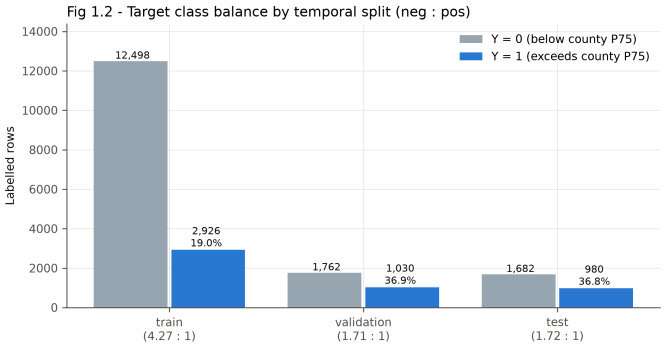


Fig. 2. Class balance by temporal split (assignment Fig. 1.2).

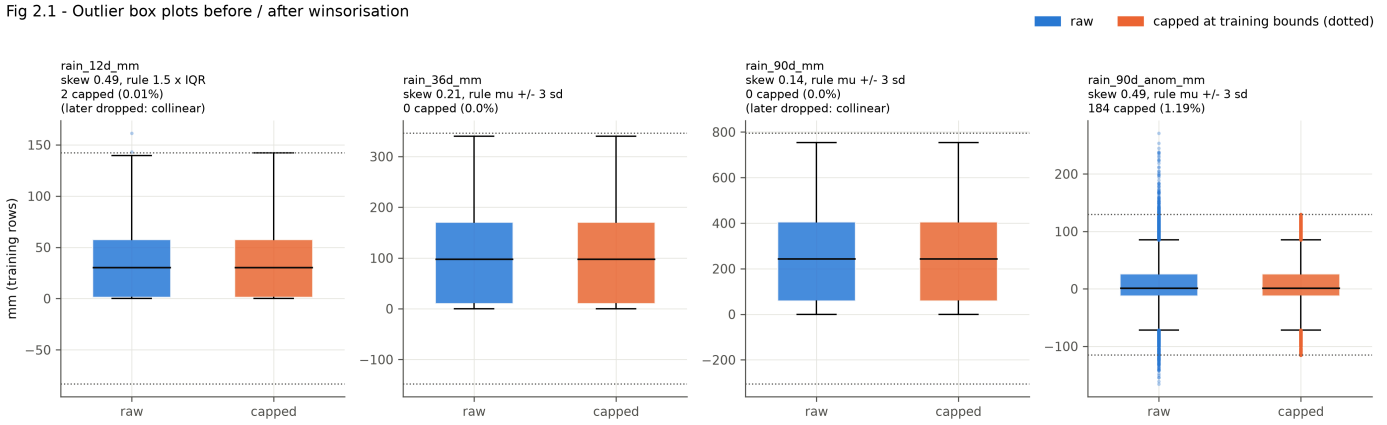


Fig. 3. Outlier box plots before and after winsorisation, training rows (assignment Fig. 2.1). Dotted lines are the training bounds.

Fig. 3. Only `rain_90d_anom_mm` has a material tail (1.19% capped). The near-normal rainfall totals fall inside $\mu \pm 3\sigma$, so capping leaves them unchanged.

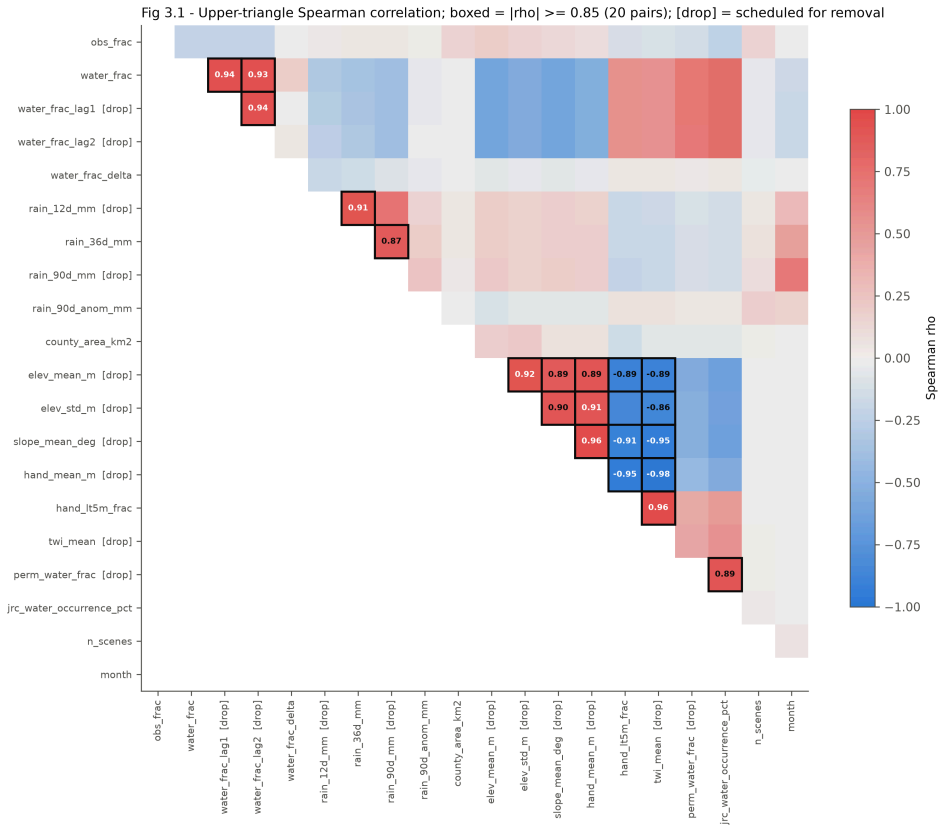


Fig. 4. Upper-triangle Spearman correlation on training rows (assignment Fig. 3.1). Boxed cells: $|\rho| \geq 0.85$; [drop]: removed by the pipeline.

Fig. 4. 20 pairs exceed the threshold. 17 of them lie inside two blocks: the water-fraction lags (3 pairs) and the terrain gradient of elevation, slope, HAND and TWI (14 pairs). The remaining 3 link the rainfall windows and the two permanent-water measures.

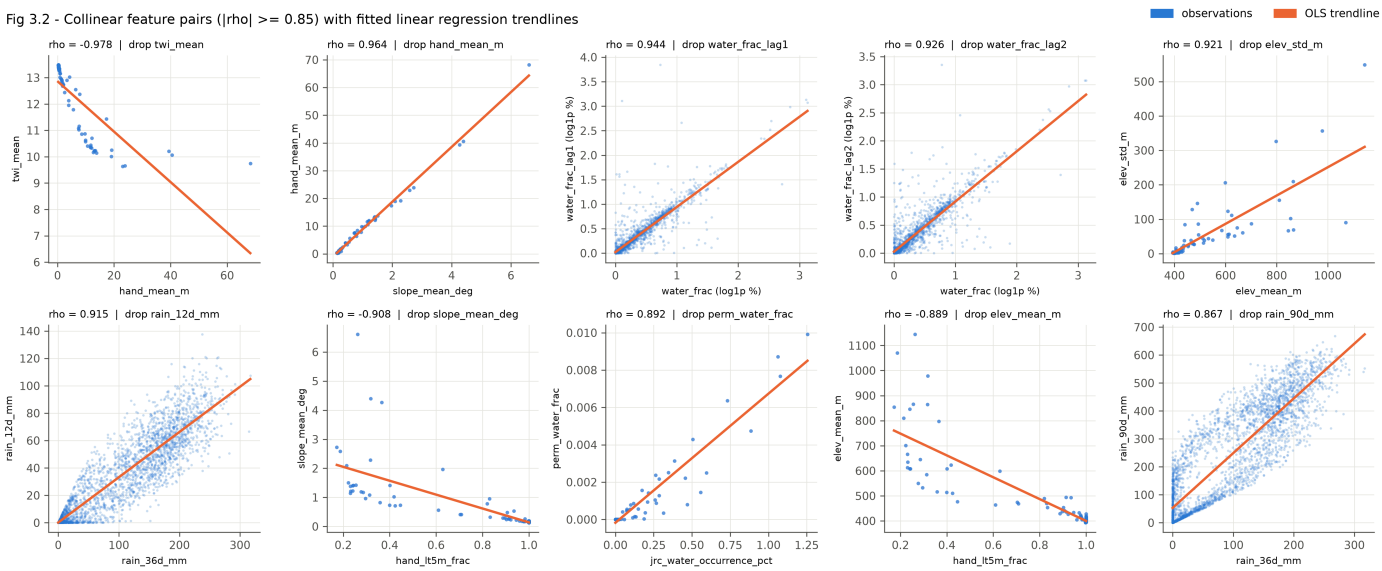


Fig. 5. Collinear pairs with fitted OLS trendlines (assignment Fig. 3.2), one panel per removed feature.

Fig. 5. Each removed feature is close to a linear function of its partner. The static panels show the 79 counties directly, which is why terrain features add little beyond one representative.

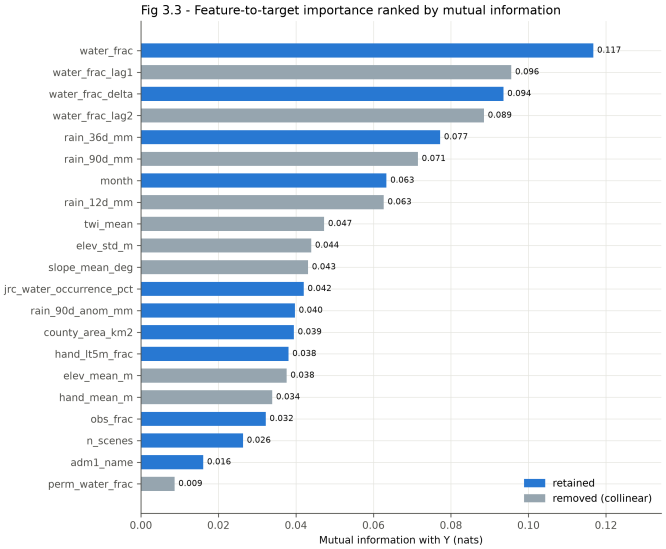


Fig. 6. Feature-to-target importance by mutual information, training rows (assignment Fig. 3.3).

Fig. 6. Current water state, momentum and recent rainfall carry the most information about Y. Several removed features rank above some retained ones; removal is driven by redundancy with a stronger partner, not by low individual signal.

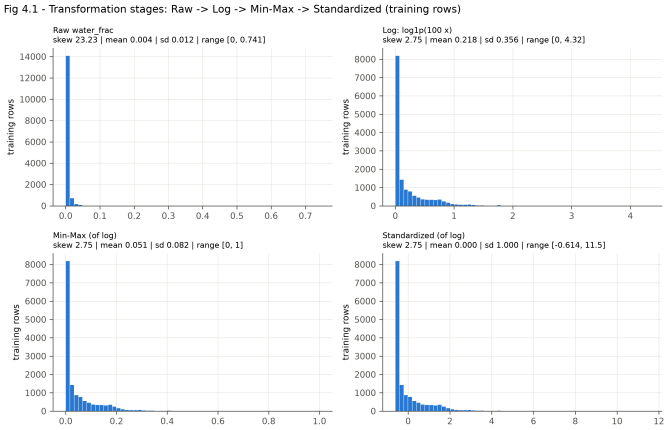


Fig. 7. Transformation stages for `water_frac`: raw, log1p(100·x), Min-Max and StandardScaler (assignment Fig. 4.1).

Fig. 7. The log step reduces skewness from 23.2 to 2.75; the distribution is far less skewed but still not symmetric. Min-Max and standardisation change scale but not shape, as expected.

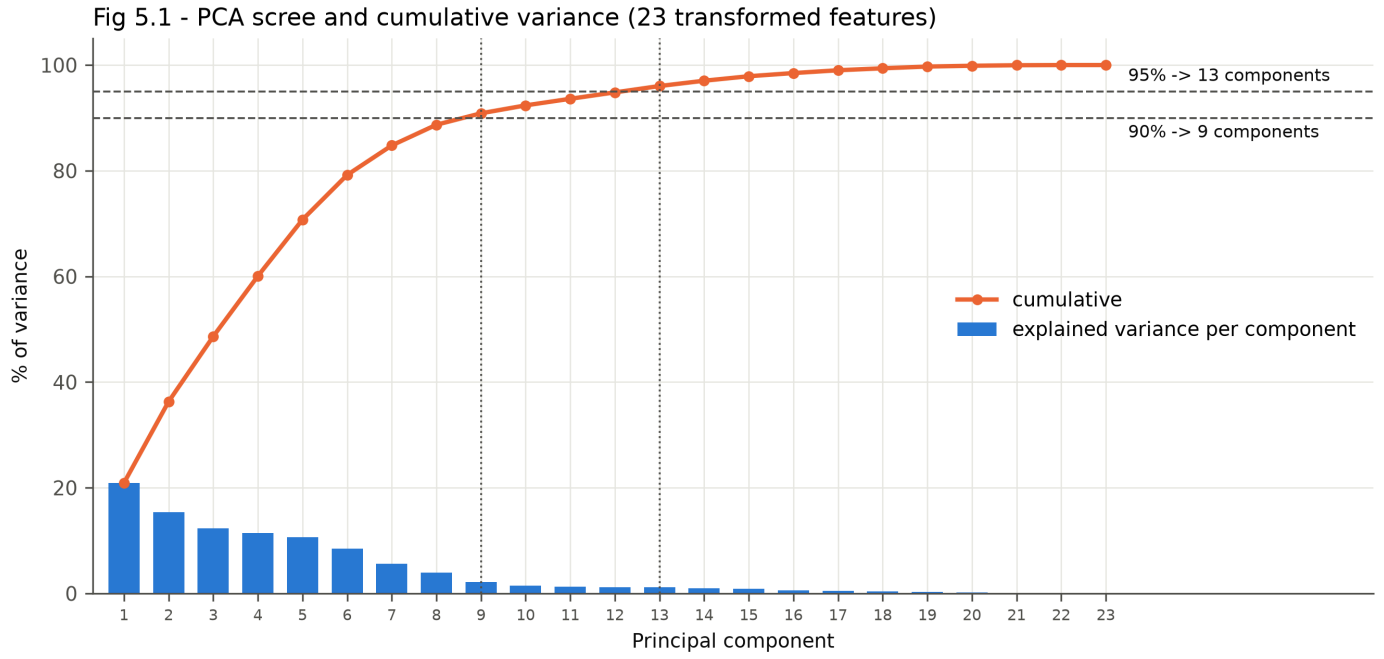


Fig. 8. PCA scree and cumulative variance on transformed training rows (assignment Fig. 5.1).

Fig. 8. 9 components explain 90% and 13 explain 95% of the variance of the 23 transformed features. No single direction dominates: the first component explains 20.92%.

IV. Decision Log

Table III compares every Phase 1 commitment with the Phase 2 implementation. "Adapted" items are explained in Section V.

Table III
Decision Log: Phase 1 Strategy versus Phase 2 Implementation

Phase 1 plan item	Status	Phase 2 implementation and evidence
Raw panel N × P and baseline missingness	As planned	Re-computed and asserted equal to Phase 1: 23,305 × 26, 1.46% missing, 19,708 complete rows.
Temporal split train < 2024-01-01 ≤ val < 2025-07-01 ≤ test	As planned	Implemented before any fitting; Table II.
Purge 2 windows at each boundary	As planned	282 rows in 4 windows removed; shares move from 73.5/13.9/12.6% to 73.9/13.4/12.8%.
Reject stratified shuffling; report class balance	As planned	Fig. 2; positive rate 19.0% train vs 36.9% / 36.8% validation / test.
Never impute Y	As planned	2,145 unlabelled rows excluded from fitting and evaluation.
County training-median imputation + indicator	As planned	`GroupMedianImputer` fitted on `X_train`; 0 NaNs remain in all three splits.
Outlier rule by skewness (Z if |skew| ≤ 0.5, else IQR)	As planned	Rules kept as pre-registered. `rain_12d_mm` skew is 0.49 on Phase 2 training rows vs 0.5 in Phase 1 (borderline); its IQR rule is kept and the feature is dropped as collinear anyway.
Cap rainfall only; never cap water fractions or static terrain	As planned	184 training values of `rain_90d_anom_mm` (1.19%) capped; `rain_36d_mm` needed none (Fig. 3).
Feature–target association by type (r_{pb}, Mann-Whitney, χ^2, MI)	As planned	MI ranking recomputed on Phase 2 training rows (Fig. 6); `water_frac` remains the strongest feature.
|ρ| ≥ 0.85 screen, lower-MI drop, VIF ≤ 10	Adapted	All 10 Phase 1 pairs reproduce (|Δρ| ≤ 0.004). Re-running the tie-break selects `twi_mean` instead of `hand_lt5m_frac` as the terrain representative (MI 0.0473 vs 0.0381); the pre-registered Phase 1 list is kept (Section V-A).
One-hot state; county not encoded; month sin/cos	As planned	11 state columns + 2 cyclical columns in the output (Section II-C).
Log + StandardScaler for skewed features; MinMax for bounded	As planned	Implemented; `water_frac` skew 23.2 → 2.75 after log (Fig. 7).
Momentum feature replacing raw lags	Adapted	Added as `MomentumFeature`: max |ρ| with retained features 0.201, MI 0.0935 (Section V-B).
`TimeSeriesSplit(5)` with 2-window gap	As planned	`temporal_cv()`, gap = 158 rows; used for the information-loss check.
`GroupKFold` by county (robustness)	As planned	PR-AUC 0.6121 ± 0.0495 on unseen counties (Section VI).
Label sensitivity: −16 dB and VH variants	As planned	27 windows re-extracted; no variant met the pre-set rule, so the Phase 1 label is kept (Table IV).
Report Sudd core separately	As planned	Core (Jonglei, Unity, Upper Nile, Lakes) within-county ρ reported in Table IV.
Check sensor effect on exceedance rate	Adapted	Implemented; result inconclusive — a monotone rise in exceedances is found instead (Section V-C).
Class-weighted estimator; validation-tuned threshold	Deferred	`class_weight="balanced"` used in the verification baseline; threshold tuning belongs to the modelling assignment.
PCA scree diagnostic	As planned	Fig. 8: 9 / 13 of 23 components for 90% / 95% variance; PCA is not applied to model inputs (Section V-D).

V. Deviations and Rationale

A. Collinearity Tie-Break Instability

The ten Phase 1 collinear pairs reproduce on the Phase 2 training rows within $|\Delta\rho| \leq 0.004$, but the lower-MI tie-break inside the terrain block is not stable. Its six features take only 79 distinct values (one per county), and their MI estimates lie between 0.0339 and 0.0473 nats, which is within estimator noise. Re-running the rule therefore keeps `twi_mean` instead of `hand_lt5m_frac`. The pipeline keeps the pre-registered Phase 1 list: changing feature choices in response to each new sample would be a form of selection leakage, and either feature represents the same highland-to-floodplain gradient.

B. Momentum Feature Added

Phase 1 proposed replacing the raw lags with a change feature. `water_frac_delta` has a maximum |ρ| of 0.201 with any retained feature, so it clears the collinearity rule, and its MI (0.0935) is close to that of the dropped lag. It is stateless and computed after imputation, so it cannot leak. Its skewness is near zero, so it is standardised without a log step.

C. Sensor Check Revealed a Rising Exceedance Rate

The wet-season (Jul–Nov) positive rate is 5.3% in 2017–2021 (two satellites), 19.1% in 2022 – mid-2025 (Sentinel-1A only) and 30.8% from mid-2025 (Sentinel-1C/D). By year it rises from 1.7% in 2019 to 16.2% in 2022 and 35.4% in 2026. The step in 2022 coincides with the loss of Sentinel-1B, but the highest rates occur after Sentinel-1C/D restored scene counts, so scene count alone does not explain the rise. It is consistent with the persistent flooding reported around Bentiu and Rubkona, which remained below the surrounding water level in October 2025 [1], but a sensor or processing contribution cannot be excluded with the data in hand. The practical consequence is non-stationarity: a threshold fixed on 2017–2023 is exceeded more often later. The modelling stage will test trend-aware features (e.g. a rolling one-year county baseline) and report performance by year.

D. Label Sensitivity and PCA Decisions

The Phase 1 label was scored against UNOSAT alongside four alternatives on the 1,657 county-date pairs (Table IV), under a rule fixed before running: adopt a variant only if median within-county ρ improves by ≥ 0.10 without pooled ρ falling by more than 0.05. None qualified, so the label is unchanged. VH < −26 dB is the most promising direction (Sudd-core ρ 0.306 vs 0.163), but the gain is below the threshold, so it is recommended for testing in the modelling stage rather than adopted now. PCA is reported as a diagnostic only: the primary model is a tree ensemble that does not need decorrelated inputs, and retaining named features keeps county forecasts explainable to humanitarian users.

TABLE IV
LABEL SENSITIVITY AGAINST UNOSAT (2022–2024)

Label variant	Pooled ρ	Median within-county ρ	Sudd-core ρ	Counties	Decision
VV −18 dB	0.696	0.064	0.163	58	kept (Phase 1)
VV −16 dB	0.617	0.065	0.111	62	no
VH −22 dB	0.523	0.008	0.052	63	no
VH −24 dB	0.612	0.079	0.121	62	no
VH −26 dB	0.678	0.129	0.306	59	no

VI. PIPELINE OUTPUT VERIFICATION

A. Dimensions

Raw panel 23,305 × 26 → 21,160 labelled rows → 20,878 after purging → train / validation / test 15,424 / 2,792 / 2,662 rows, each with 25 raw columns mapped to **23 output features**. Without the collinearity rule the same pipeline produces 34 features, so the reduction is 11 features (32%).

B. No Information Loss

Each configuration is wrapped with a class-weighted logistic regression and scored by PR-AUC under `TimeSeriesSplit(5)` with the 2-window gap on training rows, then fitted on all training rows and scored once on validation (Table V). Prevalence, the PR-AUC of a random classifier, is 0.190 (train) and 0.369 (validation).

TABLE V
INFORMATION-LOSS CHECK (TEST SET NOT USED)

Configuration	Features	CV PR-AUC (mean ± sd)	Validation PR-AUC
full (no collinearity drop)	34	0.5341 ± 0.1040	0.8305
reduced (Phase 1 drop list)	22	0.5482 ± 0.1042	0.8317
reduced + momentum (final)	23	0.5498 ± 0.1064	0.8328

Removing 12 collinear features did not reduce PR-AUC; it rose slightly (0.5341 → 0.5482 in CV, 0.8305 → 0.8317 on validation), and adding the momentum feature raised it further. The differences are much smaller than the fold-to-fold standard deviation, so the correct conclusion is that the reduction preserved information, not that it improved the model. On counties never seen during fitting (`GroupKFold`), PR-AUC is 0.6121 ± 0.0495, which supports spatial generalisation within the training period.

VII. CONCLUSION

The Phase 1 strategy was implemented as a single leakage-safe scikit-learn pipeline fitted on training rows only. It produces a 23-feature matrix with no missing values, and the reduction from 34 features lost no measurable information. Of twenty Phase 1 commitments, sixteen ran as planned, three were adapted with documented reasons and one (decision-threshold tuning) belongs to modelling. The modelling stage inherits three items: the drift in exceedance rates (Section V-C), the VH −26 dB label variant (Section V-D) and the decision threshold.

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